Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

1.1.4

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=4;i++)
4       (i-1)[arr] = (i%4);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 2;
8   }
9  int main(){
10   int arr[5] = {1,1,1,1,1};
11   func1(arr+1);
12   for(int i=0; i<5; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<5; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

0	1	2	3	4
1	1	1	1	1

3 0 2 3 0

arr[i-1] = i.1.4

Output
1 1 2 3 0
3 1 2 3 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 2;
  a = &temp;
}
```

3;

Explain your answer

Yes. As temp = arr+2, a new int variable = 3, is assigned. a, which reffered to the base address of array is now reffering to temp's address

Q. 2: Write the output of the following sequence of instructions **long arr2[60]; int arr1[50], a1=(&arr2[40]), a2=(&arr2[30]), b1=(&arr1[40]), b2=(&arr1[30]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

arr 2 ~~long~~ long

arr 1 int

~~80 80~~ 80 40 output.

"a_1" refers address of 40th element of arr 2 & "a_2" → 30th element. As arrays have continuous memory with each element occupying '8' bytes (long). Ans = 8 * (difference)

Same explanation for b1 − b2. '4' bytes.